

AnyPPG: An ECG-Guided PPG Foundation Model Trained on Over 100,000 Hours of Recordings for Holistic Health Profiling

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A Dataset Descriptions

A.1 Datasets for Pretraining

MC-MED Dataset. The MC-MED dataset [4] comprises 118,385 adult encounters recorded at the Stanford Adult Emergency Department between 2020 and 2022. It includes continuous recordings of vital signs and physiological waveforms collected from bedside monitors, such as PPG, ECG, and respiratory signals. Additionally, the dataset provides comprehensive clinical metadata, including patient demographics, medical histories, clinical orders, medication prescriptions, laboratory test results, imaging findings, and documented clinical outcomes.

PulseDB Dataset. The PulseDB dataset [15] is a curated benchmark resource designed for evaluating cuffless blood pressure estimation methods. It consists of synchronized 10-second segments of PPG, ECG, and arterial blood pressure (ABP) waveforms from 5,361 subjects, along with demographic metadata such as age. PulseDB comprises two subsets: the MIMIC-III subset ($n=2,423$), which includes recordings from patients admitted to critical care units at Beth Israel Deaconess Medical Center between 2001 and 2012 [2], and the VitalDB subset ($n=2,938$), which contains intraoperative recordings from surgical patients at Seoul National University Hospital, Republic of Korea [5].

MESA Dataset. The MESA dataset [1] is a longitudinal cohort established between 2000 and 2002 to investigate the prevalence and progression of subclinical cardiovascular disease across diverse ethnic populations. Between 2010 and 2012, 2,237 participants underwent a dedicated Sleep Exam, which included overnight polysomnography (PSG) recordings with synchronized channels such as PPG and ECG.

HSP Dataset. The HSP dataset [12] comprises 25,941 PSG recordings collected from approximately 19,492 unique patients at the Massachusetts General Hospital Sleep Laboratory as of April 1, 2023. Each recording includes a standardized set of physiological channels, with a subset featuring synchronized PPG and ECG.

CFS Dataset. The CFS dataset [8] is a longitudinal, family-based cohort designed to investigate the familial aggregation of sleep apnea. Initiated in 1990, the study enrolled 2,284 individuals (46% African American) from 361 families, with participants assessed up to four times over a 16-year follow-up period.

A.2 Datasets for Downstream Evaluation

Descriptions of the downstream datasets are provided below, with detailed statistics summarized in Table 1 and Table 2, including label distributions for both training and test splits.

PPG-DaLiA Dataset. PPG-DaLiA [9] is a multimodal dataset designed for heart rate estimation during daily life activities. It contains synchronized physiological and motion data recorded from 15 subjects performing eight distinct activities: sitting, ascending/descending stairs, table soccer, cycling, driving, lunch break, walking, and working. These activities were selected to introduce varying levels of motion artifacts under realistic conditions. The dataset includes PPG and 3D-accelerometer signals collected from both wrist- and chest-worn devices, alongside concurrent ECG recordings which serve as the ground truth.



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UCI-BP Dataset. UCI-BP [3] is a benchmark dataset for cuffless blood pressure estimation, comprising preprocessed and quality-validated recordings derived from the MIMIC-II Waveform Database. The dataset contains synchronized fingertip PPG, Lead II ECG, and invasive arterial blood pressure (ABP, in mmHg) signals sampled at 125 Hz. The invasive ABP measurements serve as reference targets for systolic and diastolic blood pressure regression.

BUT PPG Dataset. The Brno University of Technology Smartphone PPG (BUT PPG) Dataset [7] is a publicly available benchmark specifically curated for signal quality assessment and heart rate estimation on consumer devices. It comprises 3,888 ten-second PPG segments acquired from a gender-balanced cohort of 50 subjects (aged 19–76 years) under varying conditions, including rest and motion. The dataset features smartphone-recorded PPG signals (30 Hz) synchronized with high-resolution single-lead ECG (1,000 Hz) and tri-axial accelerometer data (100 Hz). Comprehensive ground truth is provided via binary signal quality labels and expert-validated heart rates derived from ECG QRS detection, alongside auxiliary physiological metadata such as blood pressure. Specifically, signal quality labels are determined by the agreement between expert-estimated heart rates from the PPG signal and an ECG-referenced heart rate. A segment is labeled as high quality if the majority of annotators provide estimates within 5 bpm of the reference, and low quality otherwise.

Gyro-Acc-PPG Dataset. Gyro-Acc-PPG contains wrist PPG recordings collected using a custom watch-type wearable device equipped with a three-channel PPG sensor, a tri-axial accelerometer, and a gyroscope. Data were acquired from 24 healthy participants (10 male and 14 female; mean age 26.9 ± 4.8 years) during a structured treadmill exercise protocol including rest, walking, and running stages. Synchronized chest ECG signals recorded with a Holter monitor serve as reference measurements for heart rate. PPG signals were sampled at 50 Hz, and ECG, accelerometer, and gyroscope signals were sampled at 125 Hz, 50 Hz, and 50 Hz, respectively.

DeepBeat Dataset. The DeepBeat dataset [13] contains PPG recordings from 132 patients with atrial fibrillation and provides publicly available validation and test splits. In our experiments, the validation split was used for model training and internal validation, while the official test split was reserved as a holdout test set. This protocol ensures strict patient-level separation between training and evaluation and enables fair performance comparison, following prior practice in [6].

WESAD Dataset. The Wearable Stress and Affect Detection (WESAD) dataset [10] is a publicly available multimodal dataset for emotion and stress recognition using wearable sensors. It contains synchronized physiological and motion recordings from 15 subjects collected during a controlled laboratory study with both chest-worn (RespiBAN) and wrist-worn (Empatica E4) devices. The chest device provides ECG sampled at 700 Hz, while the wrist device records PPG at 64 Hz. In our experiments, we consider four affective states: baseline, stress, amusement, and meditation.

Real-World PPG Dataset. The Real-World PPG dataset [11] is a publicly available dataset collected using an Internet-of-Things-based sensor platform for PPG acquisition. It contains PPG recordings from 35 healthy participants (aged 10–75 years), with 50 to 60 signal segments collected per subject. Each PPG segment consists of a 6-second recording sampled at 50 Hz, yielding 300 samples per segment. The dataset is provided with predefined training and test splits, comprising 1,374 and 700 PPG segments, respectively.

WenXinWuYang Dataset. The WenXinWuYang dataset [14] comprises physiological recordings collected using a portable device named WenXinWuYang. It includes PPG signals sampled at 25 Hz together with synchronized ECG signals, where pre-developed ECG analysis algorithms provide reference labels for downstream PPG evaluation. The dataset contains 3,345 recordings of 30-second PPG segments, which are further segmented into 10-second intervals, yielding a total of 10,035 PPG segments.

B The AnyPPG Encoder Architecture

The AnyPPG encoder is a one-dimensional convolutional neural network comprising approximately 4.04M parameters. Detailed layer-wise specifications and module definitions are provided in Table 3 and Table 4, respectively. The architecture begins with a stem layer implemented as a Type 3 ConvBlock, which consists of a 3×1 convolution followed by Batch Normalization and Swish activation, mapping the single-channel input to 64 channels. This is followed by a sequence of BasicStage modules that progressively increase the channel dimension from 64 to 512. Each stage is composed of one BasicBlock (Type 1) followed by $(n-1)$ BasicBlocks (Type 2). Internally, both block types employ a three-layer bottleneck structure consisting of a Type 1 ConvBlock (1×1 convolution), a Type 2 ConvBlock (3×1 convolution), and a second Type 1 ConvBlock (1×1 convolution). These internal ConvBlocks utilize a pre-activation design where Batch Normalization, Swish activation, and Dropout ($p = 0.5$) precede the convolution operation. A Squeeze-and-Excitation module with a reduction ratio of 2 is applied after the convolutional sequence in both block types. The BasicBlock (Type 1) is distinguished by the inclusion of a MaxPool1d layer with a kernel size of 2 and stride of 2 at the end of the module to halve the temporal resolution. The network concludes with a global mean pooling layer applied along the temporal dimension, yielding a final 512-dimensional output embedding.

C Additional Metrics for Downstream Tasks

In this section, we report evaluation metrics extending beyond the primary MAE and AUC results presented in the main text. For regression tasks, we provide the Pearson correlation coefficient (r) in Table 5. For classification tasks, we present accuracy, F1-scores and AUPRC in Tables 6, 7, and 8, respectively. Across these diverse metrics, AnyPPG consistently achieves superior performance, further validating the robustness and generalization capabilities conferred by our large-scale, multi-source pretraining strategy and ECG-guided supervision.

Table 1: Train and test label distributions of downstream datasets. For classification tasks, class counts (percentages) are reported; for regression tasks, mean (standard deviation) is reported. Abbr.: AF, atrial fibrillation; HR, heart rate; SBP/DBP, systolic/diastolic blood pressure; RR, respiratory rate.

Dataset	Task	Train	Test
Classification Tasks			
WESAD	Emotion recognition	1400 (39.4%)	346 (39.3%)
	<i>baseline, stress, amusement, meditation</i>	783 (22.0%)	196 (22.3%)
		432 (12.2%)	108 (12.3%)
		940 (26.4%)	230 (26.1%)
DeepBeat	Atrial fibrillation detection	471175 (90.8%)	13387 (76.0%)
	<i>NonAF, AF</i>	47607 (9.2%)	4230 (24.0%)
BUT PPG	Signal quality assessment	2439 (76.6%)	619 (87.7%)
	<i>Bad, Good</i>	743 (23.4%)	87 (12.3%)
Real-World PPG	Biometric identification	NA	NA
WenxinWuYang	Anomaly detection	6188 (77.1%)	1555 (77.5%)
	<i>Normal, Abnormal</i>	1840 (22.9%)	452 (22.5%)
Regression Tasks			
PPG-DaLiA	HR estimation	90.8 (23.6)	82.4 (18.5)
BUT PPG	SBP estimation	126.1 (15.2)	113.5 (12.2)
	DBP estimation	75.5 (7.1)	71.1 (10.5)
	HR estimation	75.0 (12.1)	74.9 (12.8)
Gyro-Acc-PPG	HR estimation	120.3 (24.8)	120.8 (23.1)
UCI-BP	SBP estimation	134.4 (22.9)	133.0 (23.3)
	DBP estimation	63.6 (10.7)	63.4 (10.6)
WenxinWuYang	HR estimation	80.7 (12.9)	81.1 (13.3)
	RR estimation	17.2 (2.9)	17.2 (3.0)
	Age estimation	42.8 (8.0)	43.0 (7.8)

D Ablation Experiments for AnyPPG Pretraining

In this section, we report ablation studies on two key hyperparameters in the pretraining of AnyPPG, namely batch size and temperature. As shown in Figure 1, we evaluate batch sizes of 512, 1024, 1536, 2048, and 2560, and temperature values of 0.03, 0.05, 0.07, and 0.09, where the default setting is 2560 and 0.07. Each parameter is varied independently while fixing the other to its default value. The results show that model performance remains stable across different configurations. Specifically, under varying batch sizes, the mean AUC for classification tasks ranges from 0.79 to 0.81, while the MAE for regression tasks ranges from 8.65 to 9.04. Similarly, under different temperature settings, the mean AUC ranges from 0.79 to 0.80 and the MAE ranges from 8.65 to 9.27. These results indicate that the proposed pretraining framework is not overly sensitive to these hyperparameters, demonstrating good robustness and stability in practice.

E Results for Phenome-Wide Disease Detection

This section presents detailed results of the phenome-wide disease detection evaluation. Besides, Figures 2, 3, and 4 demonstrate the subgroup analysis of model performance across age, race, and sex. The results show no substantial performance differences across sex and race groups. In contrast, the age-stratified analysis indicates that the model achieves better performance in the older subgroup (age ≥ 60) compared to the younger subgroup (age < 60), suggesting improved discriminative performance in this subgroup.

To demonstrated the specific disease phenotype with good AUC, Figure 5 displays the top-60 specific disease phenotypes ranked by AUC, excluding non-specific descriptions (e.g., those containing “other”). The results show that while the highest-performing phenotypes are concentrated in the circulatory system, effective detection extends across diverse physiological chapters.

To further characterize the decision patterns underlying these results, we analyze the embedding dimensions through the linear classification heads associated with different disease phenotypes.

Table 2: Demographic statistics of the MC-MED dataset for disease phenotype detection, aggregated at the hospital-visit level. Values are reported as mean (standard deviation) for age and as count (percentage) for categorical variables. For race, missing entries are present; therefore, counts do not sum to the total sample size.

Attribute	Train (N=59,265)	Test (N=17,995)
Age	57.9 (19.9)	47.2 (19.2)
Sex		
Male	31592 (53.3%)	10667 (59.3%)
Female	27656 (46.7%)	7323 (40.7%)
Unknown	17 (0.0%)	5 (0.0%)
Race		
White	25195 (42.5%)	6883 (38.3%)
Asian	9844 (16.6%)	3016 (16.8%)
Black or African American	4010 (6.8%)	944 (5.2%)
Native Hawaiian or Other Pacific Islander	1325 (2.2%)	285 (1.6%)
Declines to State	217 (0.0%)	116 (0.1%)
American Indian or Alaska Native	164 (0.0%)	52 (0.0%)
Other	18404 (31.1%)	6663 (37.0%)
Unknown	56 (0.0%)	26 (0.0%)

Table 3: Architecture and parameter summary of AnyPPG. C_{in} and C_{out} denote the input and output channel dimensions. Output shapes are expressed as $[B, C, L]$, where B is the batch size, C the channel dimension, and L the temporal length.

AnyPPG Encoder (Sequential Architecture)	C_{in}	C_{out}	Output Shape $[B, C, L]$	#Parameters
ConvBlock (Type 3)	1	64	$[B, 64, 1250]$	384
BasicStage ₂	64	64	$[B, 64, 625]$	32,064
BasicStage ₂	64	160	$[B, 160, 313]$	156,768
BasicStage ₂	160	160	$[B, 160, 157]$	172,320
BasicStage ₃	160	400	$[B, 400, 79]$	1,413,720
BasicStage ₃	400	400	$[B, 400, 40]$	1,510,200
BasicStage ₁	400	512	$[B, 512, 20]$	758,816
MeanPool1d	512	512	$[B, 512]$	–
Total	1	512	$[B, 512, 20]$	4,044,272

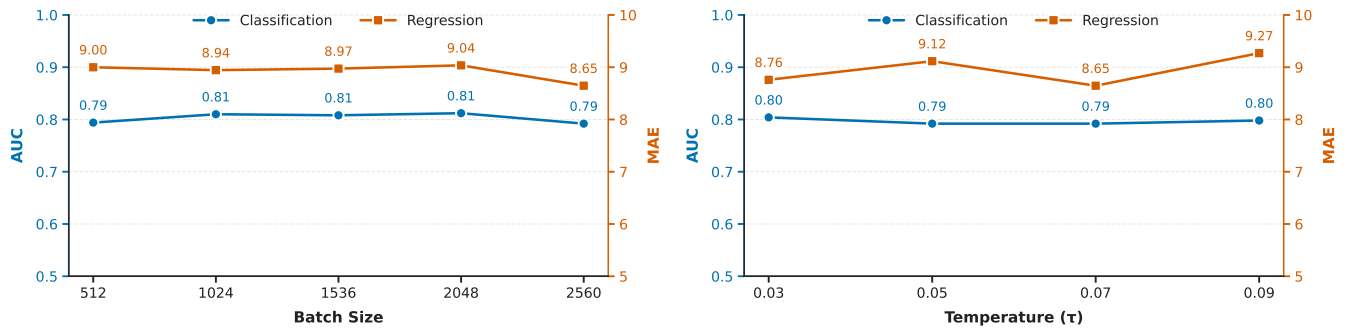


Figure 1: Ablation study on batch size and temperature in the pretraining of AnyPPG. The left panel shows the effect of varying batch size, and the right panel shows the effect of different temperature values. Performance is evaluated using mean AUC for classification tasks and MAE for regression tasks.

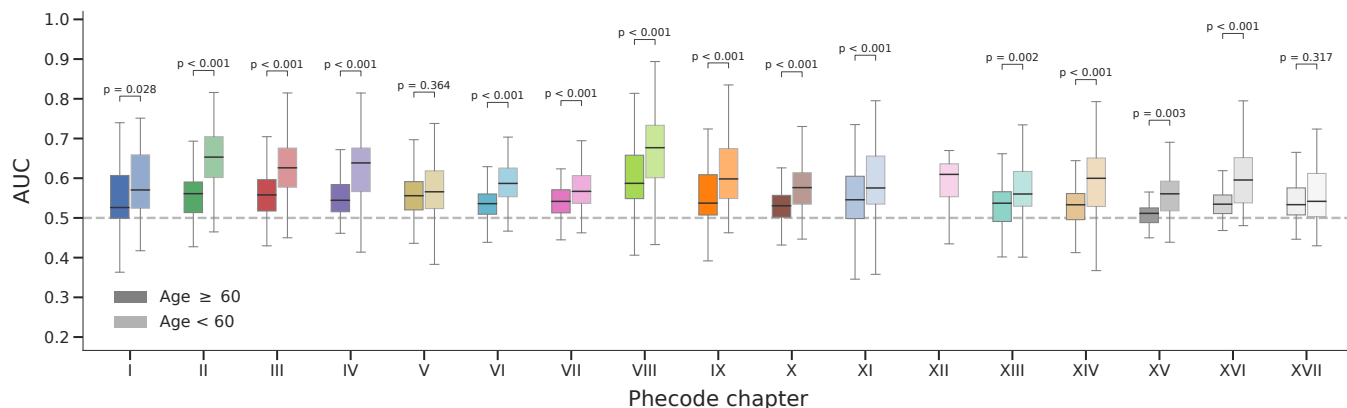


Figure 2: Subgroup analysis of phenome-wide disease detection performance stratified by age group.

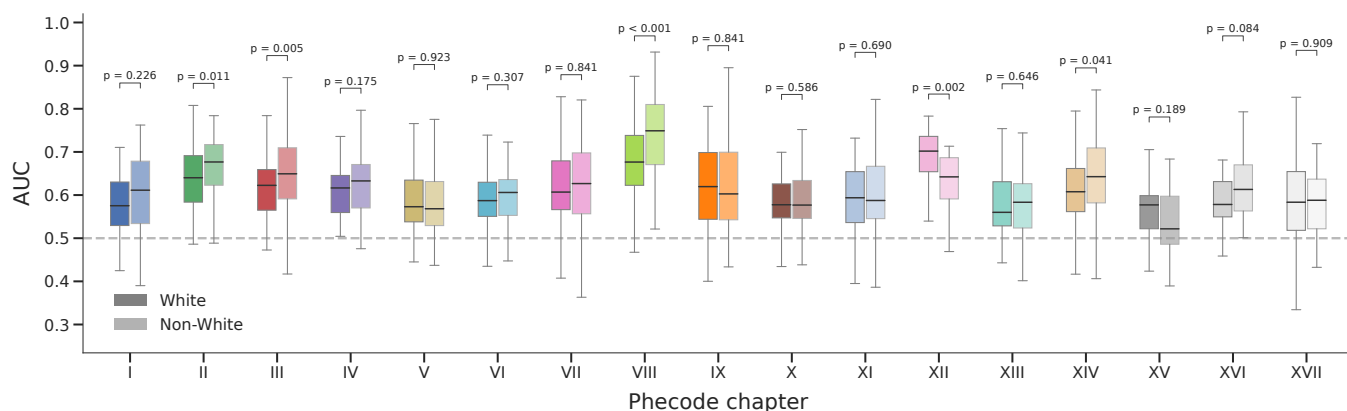


Figure 3: Subgroup analysis of phenome-wide disease detection performance stratified by race.

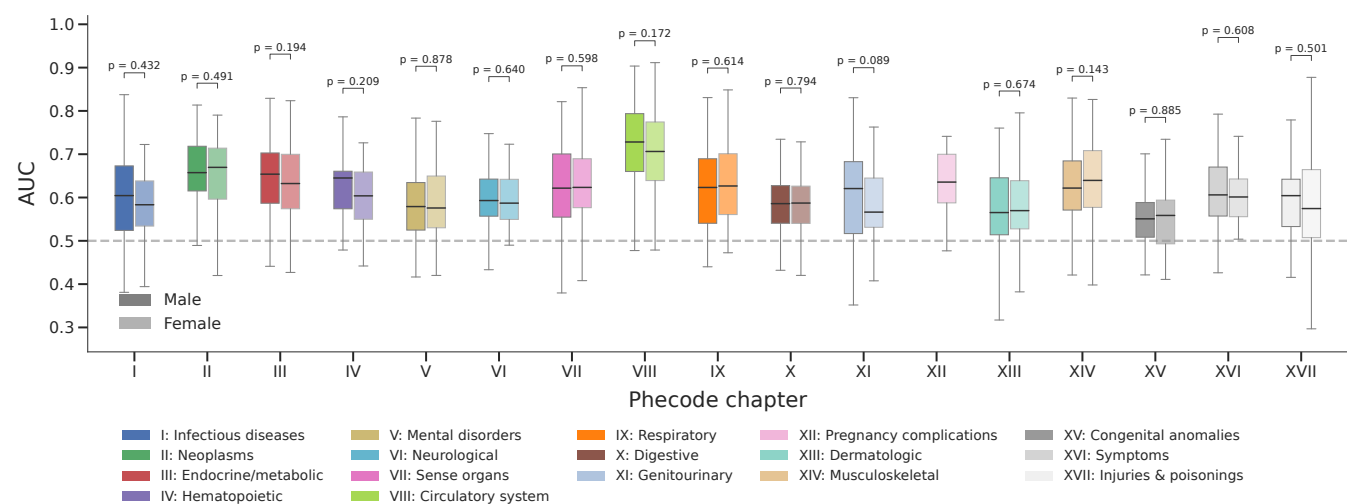


Figure 4: Subgroup analysis of phenome-wide disease detection performance stratified by sex.

Table 4: Module definitions in AnyPPG. C_{in} and C_{out} denote the input and output channel dimensions, respectively.

Module	Constituent operations (sequential definition)
$\text{BasicStage}_n(C_{in}, C_{out})$	BasicBlock (Type 1) (C_{in}, C_{out}) BasicBlock (Type 2) (C_{out}, C_{out}) $\times (n-1)$
BasicBlock (Type 1) (C_{in}, C_{out})	ConvBlock (Type 1) (C_{in}, C_{out}) ConvBlock (Type 2) (C_{out}, C_{out}) ConvBlock (Type 1) (C_{out}, C_{out}) Squeeze and excitation attention (reduction ratio=2) MaxPool1d (kernel = 2, stride = 2)
BasicBlock (Type 2) (C_{in}, C_{out})	ConvBlock (Type 1) (C_{in}, C_{out}) ConvBlock (Type 2) (C_{out}, C_{out}) ConvBlock (Type 1) (C_{out}, C_{out}) Squeeze and excitation attention (reduction ratio=2)
ConvBlock (Type 1) (C_{in}, C_{out})	BatchNorm1d(C_{in}) Swish() Dropout($p=0.5$) Conv1d(C_{in}, C_{out} , kernel_size=1, stride=1, padding=0)
ConvBlock (Type 2) (C_{in}, C_{out})	BatchNorm1d(C_{in}) Swish() Dropout($p=0.5$) Conv1d(C_{in}, C_{out} , kernel_size=3, stride=1, padding=1)
ConvBlock (Type 3) (C_{in}, C_{out})	Conv1d(C_{in}, C_{out} , kernel_size=3, stride=1, padding=1) BatchNorm1d(C_{in}) Swish()

Table 5: Linear-probing r for regression tasks. Best results are in red, second best are underlined. * indicates identical pretraining data to AnyPPG. Abbr.: HR, heart rate; SBP/DBP, systolic/diastolic blood pressure; RR, respiratory rate.

Dataset	Task	SimCLR* ICML'20	BYOL* NIPS'20	MOMENT ICML'24	Chronos-2 ArXiv'25	PulsePPG IMWUT'25	PaPaGEI-S ICLR'25	PaPaGEI-P ICLR'25	AnyPPG (Ours)
PPG-DaLiA	HR	0.76 _[0.74, 0.78]	0.76 _[0.74, 0.78]	0.73 _[0.71, 0.75]	<u>0.75</u> _[0.73, 0.77]	0.72 _[0.70, 0.74]	0.60 _[0.58, 0.63]	0.68 _[0.66, 0.70]	0.76 _[0.74, 0.78]
BUT PPG	HR	0.58 _[0.37, 0.74]	0.35 _[0.07, 0.61]	0.64 _[0.45, 0.77]	<u>0.71</u> _[0.60, 0.79]	0.57 _[0.45, 0.68]	0.45 _[0.28, 0.59]	0.07 _[-0.14, 0.39]	0.79 _[0.71, 0.86]
	SBP	-0.37 _[-0.52, -0.20]	-0.30 _[-0.49, -0.12]	<u>-0.14</u> _[-0.34, 0.04]	-0.20 _[-0.41, 0.03]	-0.32 _[-0.49, -0.12]	-0.17 _[-0.40, 0.02]	-0.19 _[-0.30, -0.07]	0.04 _[-0.20, 0.28]
	DBP	-0.24 _[-0.41, -0.04]	-0.23 _[-0.35, -0.12]	<u>-0.18</u> _[-0.39, 0.06]	0.00 _[-0.21, 0.21]	-0.21 _[-0.41, 0.03]	-0.28 _[-0.48, -0.07]	0.26 _[0.08, 0.43]	0.22 _[0.03, 0.40]
Gyro-Acc-PPG	HR	<u>0.74</u> _[0.68, 0.79]	0.73 _[0.66, 0.78]	0.65 _[0.58, 0.72]	0.69 _[0.64, 0.74]	0.68 _[0.62, 0.74]	0.39 _[0.30, 0.47]	0.60 _[0.53, 0.66]	0.76 _[0.70, 0.81]
UCI-BP	SBP	0.52 _[0.51, 0.52]	<u>0.50</u> _[0.49, 0.51]	0.45 _[0.44, 0.46]	0.39 _[0.39, 0.40]	0.45 _[0.45, 0.46]	0.34 _[0.33, 0.35]	0.42 _[0.41, 0.43]	0.52 _[0.51, 0.53]
	DBP	0.44 _[0.43, 0.45]	0.41 _[0.40, 0.41]	0.35 _[0.34, 0.36]	0.30 _[0.29, 0.30]	0.34 _[0.33, 0.34]	0.21 _[0.21, 0.22]	0.31 _[0.30, 0.32]	<u>0.43</u> _[0.42, 0.44]
WenXinWuYang	HR	0.62 _[0.58, 0.65]	0.62 _[0.59, 0.65]	0.62 _[0.58, 0.65]	0.62 _[0.58, 0.65]	0.60 _[0.56, 0.63]	0.35 _[0.31, 0.40]	0.52 _[0.48, 0.55]	0.63 _[0.60, 0.67]
	RR	<u>0.13</u> _[0.09, 0.18]	0.12 _[0.08, 0.16]	0.10 _[0.06, 0.14]	0.13 _[0.09, 0.17]	0.08 _[0.04, 0.13]	0.07 _[0.03, 0.11]	0.12 _[0.08, 0.16]	0.17 _[0.13, 0.20]
	Age	0.29 _[0.25, 0.33]	<u>0.31</u> _[0.26, 0.34]	0.26 _[0.22, 0.30]	0.26 _[0.22, 0.30]	0.23 _[0.18, 0.27]	0.12 _[0.08, 0.17]	0.21 _[0.17, 0.25]	0.33 _[0.29, 0.36]

Specifically, we extract the weight vectors (512-dimensional) from the linear classification heads corresponding to each phenotype and compute the normalized average weights across circulatory and non-circulatory disease categories (with at least 100 positive samples and an AUC ≥ 0.70) to quantify the relative importance of embedding dimensions (Figure 6). The analysis reveals that certain dimensions exhibit consistently high importance across both categories (highlighted by green arrows), suggesting shared physiological factors. In contrast, other dimensions show category-specific

prominence, with representative examples highlighted by brown arrows, illustrating dimensions that are preferentially emphasized in different disease categories. These patterns indicate that the model captures both shared and category-specific decision structures rather than relying on a fixed set of generic features.

Finally, Table 9 provides the complete list of all 307 phenotypes that achieved an AUC of ≥ 0.70 , along with the corresponding 95% confidence intervals and FDR-corrected p-values. In addition, we report AUPRC lift, defined as AUPRC normalized by disease

Table 6: Linear-probing ACC for classification downstream tasks. Best results are in red, second best are underlined. * indicates identical pretraining data to AnyPPG.

Dataset	Task	SimCLR* ICML'20	BYOL* NIPS'20	MOMENT ICML'24	Chronos-2 ArXiv'25	PulsePPG IMWUT'25	PaPaGEI-S ICLR'25	PaPaGEI-P ICLR'25	AnyPPG (Ours)
WESAD	Emotion recognition	0.62 _[0.58, 0.65]	0.57 _[0.54, 0.60]	0.54 _[0.51, 0.57]	0.55 _[0.52, 0.58]	0.56 _[0.53, 0.59]	0.47 _[0.44, 0.51]	0.54 _[0.51, 0.57]	<u>0.61</u> _[0.58, 0.64]
DeepBeat	Atrial fibrillation detection	0.77 _[0.77, 0.78]	<u>0.76</u> _[0.76, 0.77]	<u>0.76</u> _[0.75, 0.76]	<u>0.76</u> _[0.75, 0.76]	0.75 _[0.75, 0.76]	<u>0.76</u> _[0.75, 0.77]	<u>0.76</u> _[0.75, 0.76]	<u>0.76</u> _[0.76, 0.77]
BUT PPG	Signal quality assessment	0.86 _[0.83, 0.88]	0.86 _[0.84, 0.89]	0.86 _[0.84, 0.89]	0.86 _[0.84, 0.89]	0.86 _[0.83, 0.88]	0.86 _[0.83, 0.89]	<u>0.87</u> _[0.84, 0.89]	0.88 _[0.85, 0.90]
Real-World PPG	Biometric identification	0.98 _[0.97, 0.99]	<u>0.93</u> _[0.91, 0.95]	0.61 _[0.57, 0.64]	0.37 _[0.33, 0.40]	0.89 _[0.87, 0.91]	0.33 _[0.29, 0.36]	0.84 _[0.81, 0.86]	0.98 _[0.97, 0.99]
WenXinWuYang	Anomaly detection	0.79 _[0.78, 0.81]	0.80 _[0.79, 0.82]	0.80 _[0.79, 0.82]	<u>0.81</u> _[0.79, 0.82]	0.80 _[0.79, 0.82]	0.77 _[0.76, 0.79]	0.80 _[0.78, 0.82]	0.83 _[0.81, 0.84]

Table 7: Linear-probing F1-score for classification downstream tasks. Best results are in red, second best are underlined. * indicates identical pretraining data to AnyPPG.

Dataset	Task	SimCLR* ICML'20	BYOL* NIPS'20	MOMENT ICML'24	Chronos-2 ArXiv'25	PulsePPG IMWUT'25	PaPaGEI-S ICLR'25	PaPaGEI-P ICLR'25	AnyPPG (Ours)
WESAD	Emotion recognition	0.51 _[0.48, 0.54]	0.47 _[0.44, 0.49]	0.43 _[0.41, 0.46]	0.44 _[0.42, 0.47]	0.44 _[0.42, 0.47]	0.37 _[0.35, 0.40]	0.42 _[0.40, 0.45]	<u>0.50</u> _[0.48, 0.53]
DeepBeat	Atrial fibrillation detection	<u>0.55</u> _[0.54, 0.56]	0.52 _[0.51, 0.53]	0.51 _[0.50, 0.51]	0.57 _[0.56, 0.58]	0.47 _[0.46, 0.47]	0.43 _[0.43, 0.43]	0.43 _[0.43, 0.44]	0.57 _[0.56, 0.58]
BUT PPG	Signal quality assessment	0.59 _[0.54, 0.64]	0.59 _[0.54, 0.64]	<u>0.60</u> _[0.55, 0.66]	0.59 _[0.54, 0.64]	0.55 _[0.51, 0.60]	0.54 _[0.50, 0.59]	0.59 _[0.54, 0.64]	0.67 _[0.62, 0.73]
Real-World PPG	Biometric identification	0.97 _[0.96, 0.99]	<u>0.92</u> _[0.90, 0.94]	0.57 _[0.53, 0.61]	0.34 _[0.30, 0.37]	0.88 _[0.85, 0.90]	0.29 _[0.26, 0.32]	0.82 _[0.79, 0.84]	0.97 _[0.96, 0.99]
WenXinWuYang	Anomaly detection	0.57 _[0.54, 0.59]	0.58 _[0.55, 0.61]	<u>0.61</u> _[0.58, 0.63]	0.60 _[0.57, 0.63]	<u>0.61</u> _[0.59, 0.64]	0.44 _[0.43, 0.44]	0.59 _[0.56, 0.61]	0.67 _[0.65, 0.70]

Table 8: Linear-probing AUPRC for classification downstream tasks. Best results are in red, second best are underlined. * indicates identical pretraining data to AnyPPG.

Dataset	Task	SimCLR* ICML'20	BYOL* NIPS'20	MOMENT ICML'24	Chronos-2 ArXiv'25	PulsePPG IMWUT'25	PaPaGEI-S ICLR'25	PaPaGEI-P ICLR'25	AnyPPG (Ours)
WESAD	Emotion recognition	<u>0.55</u> _[0.52, 0.58]	0.52 _[0.49, 0.55]	0.47 _[0.45, 0.50]	0.50 _[0.47, 0.53]	0.50 _[0.47, 0.52]	0.41 _[0.39, 0.44]	0.49 _[0.46, 0.51]	0.58 _[0.56, 0.61]
DeepBeat	Atrial fibrillation detection	0.43 _[0.41, 0.44]	0.37 _[0.36, 0.39]	0.33 _[0.32, 0.35]	<u>0.42</u> _[0.40, 0.43]	0.30 _[0.29, 0.31]	0.27 _[0.26, 0.28]	0.27 _[0.26, 0.28]	0.41 _[0.40, 0.43]
BUT PPG	Signal quality assessment	0.31 _[0.24, 0.40]	<u>0.35</u> _[0.26, 0.44]	0.32 _[0.24, 0.41]	0.34 _[0.26, 0.43]	0.29 _[0.22, 0.37]	0.25 _[0.18, 0.34]	0.29 _[0.22, 0.38]	0.38 _[0.28, 0.47]
Real-World PPG	Biometric identification	0.99 _[0.98, 1.00]	<u>0.97</u> _[0.96, 0.98]	0.64 _[0.61, 0.67]	0.38 _[0.35, 0.41]	0.94 _[0.93, 0.96]	0.34 _[0.31, 0.37]	0.89 _[0.87, 0.91]	0.99 _[0.99, 1.00]
WenXinWuYang	Anomaly detection	0.45 _[0.40, 0.49]	0.47 _[0.42, 0.51]	0.48 _[0.44, 0.53]	<u>0.51</u> _[0.47, 0.55]	0.48 _[0.43, 0.52]	0.27 _[0.24, 0.29]	0.45 _[0.41, 0.50]	0.56 _[0.51, 0.60]

prevalence, to facilitate comparison across phenotypes with varying prevalence and to provide a more interpretable measure of performance relative to the random baseline.

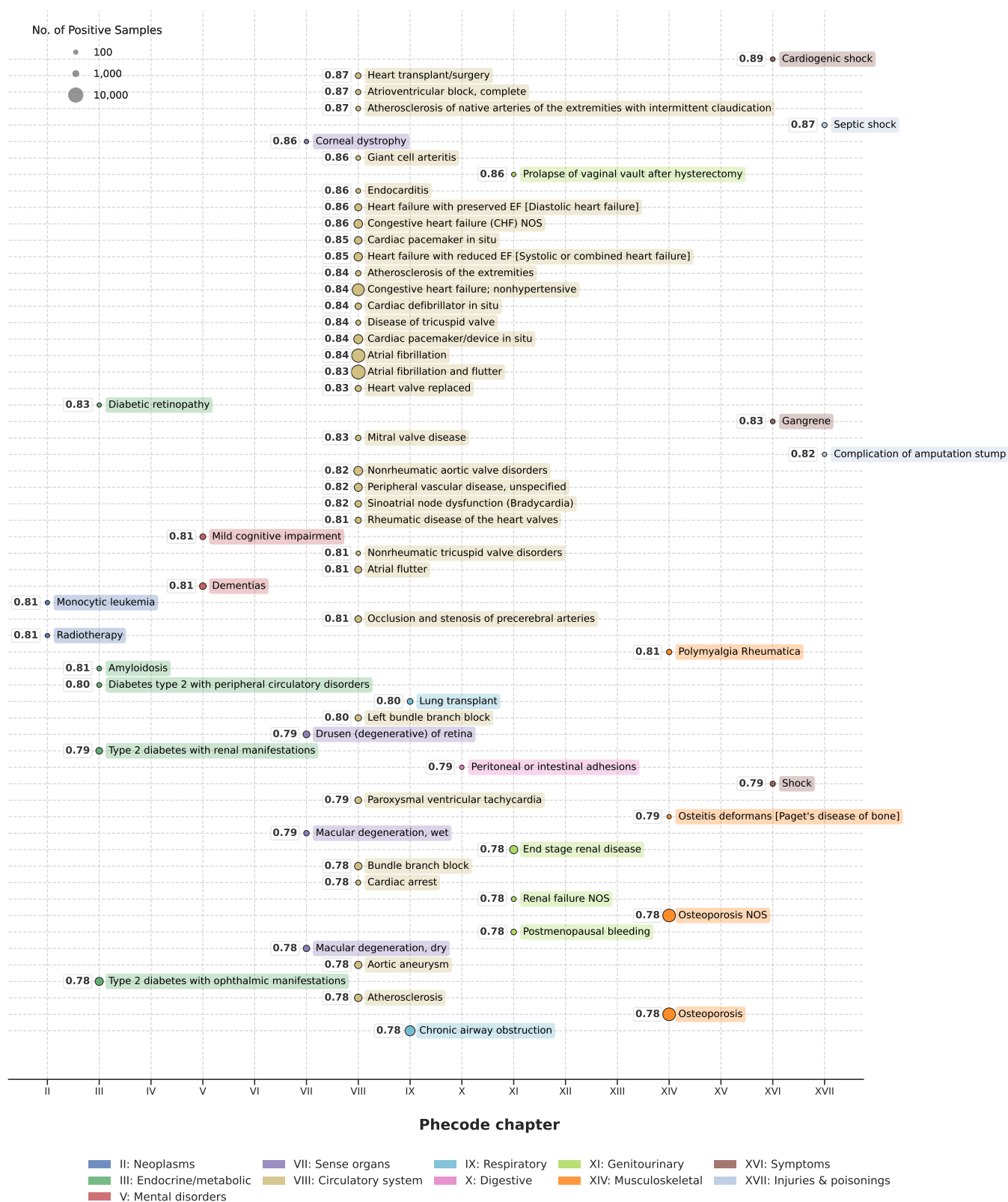


Figure 5: The top 60 disease phenotypes with the highest discriminative performance, ranked by AUC.

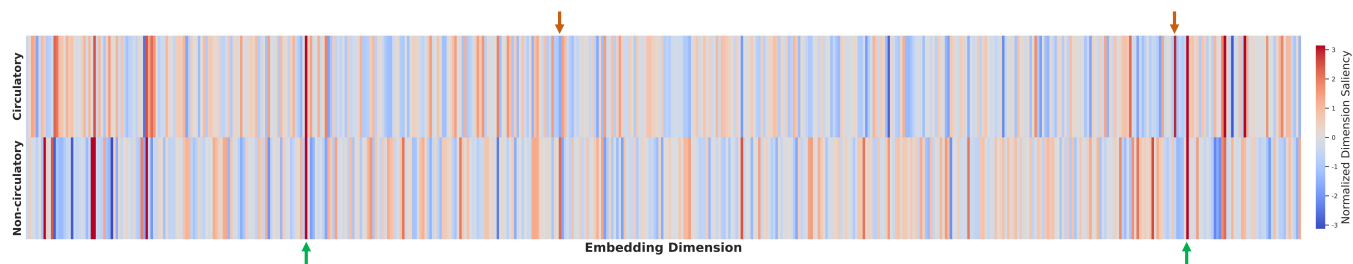


Figure 6: Embedding dimension analysis based on linear classification heads. The figure shows the normalized embedding saliency across dimensions, where higher values indicate greater importance for disease detection. Results are aggregated over disease phenotypes within circulatory and non-circulatory categories. Green arrows mark dimensions with high saliency in both categories, while brown arrows highlight representative dimensions with differing saliency between the two categories.

Table 9: Detailed phenome-wide disease detection results. N_{pos} and N_{neg} denote the numbers of positive and negative samples, respectively. AUC denotes the area under the ROC curve. P-value corresponds to the statistical significance of the AUC. AUPRC lift is defined as AUPRC normalized by disease prevalence ($N_{pos}/(N_{pos} + N_{neg})$).

Phecode	Description	N_{pos}	N_{neg}	AUC	P-Value	AUPRC Lift
Chapter I: Infectious diseases						
38.1	Gram negative septicemia	280	359620	0.74 (0.71, 0.77)	<0.001	2.88
41.2	Streptococcus infection	360	359540	0.73 (0.70, 0.75)	<0.001	2.37
117.4	Aspergillosis	180	359720	0.72 (0.69, 0.75)	<0.001	2.05
38	Septicemia	6920	352980	0.72 (0.71, 0.72)	<0.001	2.68
41.12	Methicillin resistant Staphylococcus aureus	640	359260	0.70 (0.68, 0.72)	<0.001	1.91
Chapter II: Neoplasms						
204.3	Monocytic leukemia	100	359800	0.81 (0.78, 0.84)	<0.001	2.83
196	Radiotherapy	100	359800	0.81 (0.78, 0.84)	<0.001	10.4
189.4	Malignant neoplasm of other urinary organs	360	359540	0.79 (0.77, 0.80)	<0.001	2.89
172.22	Squamous cell carcinoma	1320	358580	0.77 (0.76, 0.78)	<0.001	2.87
189.2	Cancer of bladder	1200	358700	0.75 (0.74, 0.76)	<0.001	2.56
172.3	Carcinoma in situ of skin	560	359340	0.75 (0.73, 0.76)	<0.001	2.12
198.6	Secondary malignancy of bone	2140	357760	0.75 (0.74, 0.75)	<0.001	4.42
198.2	Secondary malignancy of respiratory organs	780	359120	0.74 (0.73, 0.76)	<0.001	3.95
204.21	Myeloid leukemia, acute	560	359340	0.73 (0.71, 0.75)	<0.001	2.65
172.2	Other non-epithelial cancer of skin	6040	353860	0.73 (0.73, 0.74)	<0.001	2.37
155	Cancer of liver and intrahepatic bile duct	1460	358440	0.73 (0.72, 0.74)	<0.001	2.21
195	Cancer, suspected or other	4100	355800	0.73 (0.72, 0.74)	<0.001	3.37
189.21	Malignant neoplasm of bladder	1100	358800	0.73 (0.72, 0.74)	<0.001	2.16
165.1	Cancer of bronchus; lung	2080	357820	0.73 (0.72, 0.74)	<0.001	2.17
165	Cancer within the respiratory system	2100	357800	0.73 (0.72, 0.73)	<0.001	2.11
202.22	Reticulosarcoma	1020	358880	0.73 (0.71, 0.74)	<0.001	2.76
185	Cancer of prostate	3640	356260	0.72 (0.72, 0.73)	<0.001	2.22
155.1	Malignant neoplasm of liver, primary	1060	358840	0.72 (0.71, 0.74)	<0.001	2.13
159.3	Malignant neoplasm of gallbladder and extrahepatic bile ducts	240	359660	0.72 (0.70, 0.75)	<0.001	2.11
198	Secondary malignant neoplasm	5580	354320	0.72 (0.72, 0.73)	<0.001	2.85
172.21	Basal cell carcinoma	3300	356600	0.72 (0.71, 0.73)	<0.001	2.15
198.5	Secondary malignancy of brain/spine	1040	358860	0.72 (0.71, 0.73)	<0.001	2.32
159.2	Malignant neoplasm of small intestine, including duodenum	200	359700	0.72 (0.69, 0.75)	<0.001	2.28
174.1	Breast cancer [female]	7060	352840	0.72 (0.71, 0.72)	<0.001	2.18
172	Skin cancer	7600	352300	0.72 (0.71, 0.72)	<0.001	2.33
189	Cancer of urinary organs (incl. kidney and bladder)	2480	357420	0.72 (0.71, 0.73)	<0.001	2.26
182	Malignant neoplasm of uterus	1200	358700	0.72 (0.70, 0.73)	<0.001	2.42
149.1	Cancer of oropharynx	380	359520	0.71 (0.70, 0.73)	<0.001	1.85
204.4	Multiple myeloma	940	358960	0.71 (0.70, 0.73)	<0.001	1.98
174	Breast cancer	7140	352760	0.71 (0.71, 0.72)	<0.001	2.06
172.1	Melanomas of skin, dx or hx	3700	356200	0.71 (0.71, 0.72)	<0.001	2.39
184.1	Malignant neoplasm of ovary and other uterine adnexa	1060	358840	0.71 (0.70, 0.72)	<0.001	2.56
195.1	Malignant neoplasm, other	2060	357840	0.71 (0.70, 0.72)	<0.001	2.82
149.4	Cancer of larynx	160	359740	0.71 (0.68, 0.73)	<0.001	1.71
174.11	Malignant neoplasm of female breast	6460	353440	0.71 (0.70, 0.71)	<0.001	2.00
159	Malignant neoplasm of other and ill-defined sites within the digestive organs and peritoneum	900	359000	0.71 (0.69, 0.72)	<0.001	2.40
204.2	Myeloid leukemia	920	358980	0.70 (0.69, 0.72)	<0.001	2.09
151	Cancer of stomach	560	359340	0.70 (0.68, 0.72)	<0.001	2.31
184	Cancer of other female genital organs	1660	358240	0.70 (0.69, 0.71)	<0.001	2.76
Chapter III: Endocrine/metabolic						
250.7	Diabetic retinopathy	100	359800	0.83 (0.80, 0.86)	<0.001	4.41
270.33	Amyloidosis	300	359600	0.81 (0.78, 0.83)	<0.001	7.81
250.25	Diabetes type 2 with peripheral circulatory disorders	380	359520	0.80 (0.78, 0.82)	<0.001	3.36
250.22	Type 2 diabetes with renal manifestations	1480	358420	0.79 (0.78, 0.80)	<0.001	4.60
250.23	Type 2 diabetes with ophthalmic manifestations	2600	357300	0.78 (0.77, 0.79)	<0.001	4.03
250.11	Type 1 diabetes with ketoacidosis	240	359660	0.77 (0.75, 0.79)	<0.001	2.50

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Table 9 continued from previous page

Phecode	Description	N_{pos}	N_{neg}	AUC	P-Value	AUPRC Lift
270.32	Paraproteinemia	720	359180	0.77 (0.75, 0.78)	<0.001	2.62
260.3	Adult failure to thrive	1740	358160	0.76 (0.75, 0.77)	<0.001	3.37
276.13	Hyperpotassemia	1620	358280	0.75 (0.74, 0.76)	<0.001	2.83
253.7	Other disorders of neurohypophysis	180	359720	0.75 (0.72, 0.78)	<0.001	2.59
274.2	Crystal arthropathies	280	359620	0.74 (0.72, 0.76)	<0.001	1.91
276.6	Fluid overload	1180	358720	0.74 (0.73, 0.75)	<0.001	2.74
274.21	Chondrocalcinosis	280	359620	0.73 (0.72, 0.76)	<0.001	1.95
275.3	Disorders of magnesium metabolism	1000	358900	0.73 (0.71, 0.74)	<0.001	2.51
250.24	Type 2 diabetes with neurological manifestations	1660	358240	0.72 (0.71, 0.73)	<0.001	2.09
276.11	Hyperosmolality and/or hyponatremia	200	359700	0.72 (0.70, 0.75)	<0.001	2.02
250.2	Type 2 diabetes	23580	336320	0.72 (0.72, 0.73)	<0.001	2.12
255.12	Hyperaldosteronism	120	359780	0.72 (0.68, 0.76)	<0.001	2.56
270.11	Disturbances of sulphur-bearing amino-acid metabolism	200	359700	0.72 (0.69, 0.75)	<0.001	1.98
249	Secondary diabetes mellitus	740	359160	0.71 (0.70, 0.73)	<0.001	2.40
276.12	Hyposmolality and/or hyponatremia	5580	354320	0.71 (0.71, 0.72)	<0.001	2.19
275.6	Hypercalcemia	1320	358580	0.71 (0.70, 0.72)	<0.001	2.06
276.4	Acid-base balance disorder	1680	358220	0.71 (0.70, 0.72)	<0.001	2.41
250.3	Insulin pump user	1600	358300	0.71 (0.70, 0.72)	<0.001	2.31
250	Diabetes mellitus	31380	328520	0.71 (0.70, 0.71)	<0.001	1.93
276.41	Acidosis	1640	358260	0.70 (0.69, 0.71)	<0.001	2.42
251.1	Hypoglycemia	740	359160	0.70 (0.68, 0.72)	<0.001	3.22
Chapter IV: Hematopoietic						
285.21	Anemia in chronic kidney disease	1260	358640	0.75 (0.74, 0.76)	<0.001	5.16
286.2	Encounter for long-term (current) use of anticoagulants	3760	356140	0.75 (0.74, 0.76)	<0.001	3.56
286.9	Abnormal coagulation profile	400	359500	0.74 (0.71, 0.76)	<0.001	4.62
285.2	Anemia of chronic disease	2220	357680	0.72 (0.71, 0.73)	<0.001	2.60
286.6	Defibrination syndrome	160	359740	0.71 (0.66, 0.75)	<0.001	2.87
281.12	Other vitamin B12 deficiency anemia	480	359420	0.71 (0.68, 0.73)	<0.001	2.16
Chapter V: Mental disorders						
292.2	Mild cognitive impairment	900	359000	0.81 (0.80, 0.83)	<0.001	4.59
290.1	Dementias	1480	358420	0.81 (0.80, 0.82)	<0.001	3.73
315.3	Intellectual disability	180	359720	0.77 (0.74, 0.80)	<0.001	3.38
290.11	Alzheimer's disease	140	359760	0.76 (0.73, 0.78)	<0.001	2.06
290	Delirium dementia and amnestic and other cognitive disorders	2600	357300	0.75 (0.74, 0.76)	<0.001	2.63
292.11	Aphasia	600	359300	0.72 (0.70, 0.73)	<0.001	2.00
292.4	Altered mental status	5000	354900	0.71 (0.70, 0.72)	<0.001	2.33
292.3	Memory loss	2400	357500	0.70 (0.69, 0.71)	<0.001	1.87
Chapter VI: Neurological						
332	Parkinson's disease	1300	358600	0.77 (0.76, 0.78)	<0.001	2.75
350.5	Abnormal reflex	160	359740	0.74 (0.70, 0.77)	<0.001	3.21
327.5	Parasomnia	360	359540	0.73 (0.70, 0.75)	<0.001	2.23
359.2	Myopathy	260	359640	0.71 (0.67, 0.74)	<0.001	3.04
350.2	Abnormality of gait	7780	352120	0.71 (0.70, 0.71)	<0.001	2.03
356	Hereditary and idiopathic peripheral neuropathy	560	359340	0.70 (0.69, 0.72)	<0.001	1.86
333.1	Essential tremor	780	359120	0.70 (0.68, 0.72)	<0.001	2.04
Chapter VII: Sense organs						
364.5	Corneal dystrophy	120	359780	0.86 (0.84, 0.89)	<0.001	9.90
362.27	Drusen (degenerative) of retina	1560	358340	0.79 (0.78, 0.80)	<0.001	3.24
362.22	Macular degeneration, wet	680	359220	0.79 (0.77, 0.80)	<0.001	2.44
362.21	Macular degeneration, dry	1320	358580	0.78 (0.77, 0.79)	<0.001	2.84
362.2	Degeneration of macula and posterior pole of retina	5840	354060	0.77 (0.76, 0.77)	<0.001	2.92
362.29	Macular degeneration (senile) of retina NOS	640	359260	0.77 (0.75, 0.79)	<0.001	3.12
366	Cataract	17760	342140	0.77 (0.76, 0.77)	<0.001	2.55
362.26	Macular puckering of retina	2540	357360	0.76 (0.76, 0.77)	<0.001	2.81
366.2	Senile cataract	11200	348700	0.75 (0.75, 0.76)	<0.001	2.41
365.11	Primary open angle glaucoma	1400	358500	0.75 (0.74, 0.76)	<0.001	2.75
365.1	Open-angle glaucoma	1920	357980	0.75 (0.74, 0.76)	<0.001	2.46

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Table 9 continued from previous page

Phecode	Description	N_{pos}	N_{neg}	AUC	P-Value	AUPRC Lift
374.6	Dermatochalasis	1020	358880	0.75 (0.73, 0.76)	<0.001	2.63
374.1	Ectropion or entropion	380	359520	0.74 (0.72, 0.76)	<0.001	2.20
389.1	Sensorineural hearing loss	4540	355360	0.74 (0.73, 0.74)	<0.001	2.32
365.2	Primary angle-closure glaucoma	1700	358200	0.73 (0.72, 0.74)	<0.001	2.72
367.4	Presbyopia	4200	355700	0.72 (0.72, 0.73)	<0.001	2.09
365	Glaucoma	8360	351540	0.72 (0.72, 0.73)	<0.001	2.33
379.5	Disorders of iris and ciliary body	340	359560	0.72 (0.70, 0.75)	<0.001	2.66
364.2	Corneal edema	180	359720	0.72 (0.68, 0.76)	<0.001	3.21
362	Other retinal disorders	9280	350620	0.72 (0.71, 0.72)	<0.001	2.38
362.23	Cystoid macular degeneration of retina	1300	358600	0.72 (0.70, 0.73)	<0.001	2.25
389.3	Degenerative and vascular disorders of ear	520	359380	0.72 (0.70, 0.73)	<0.001	1.87
375	Disorders of lacrimal system	4820	355080	0.71 (0.71, 0.72)	<0.001	2.33
375.1	Dry eyes	4500	355400	0.71 (0.70, 0.72)	<0.001	2.29
389.5	Disorders of acoustic nerve	160	359740	0.70 (0.67, 0.73)	<0.001	1.84
Chapter VIII: Circulatory system						
429.1	Heart transplant/surgery	800	359100	0.87 (0.86, 0.88)	<0.001	5.48
426.24	Atrioventricular block, complete	520	359380	0.87 (0.86, 0.88)	<0.001	12.85
440.22	Atherosclerosis of native arteries of the extremities with intermittent claudication	300	359600	0.87 (0.85, 0.88)	<0.001	6.03
446.5	Giant cell arteritis	260	359640	0.86 (0.84, 0.88)	<0.001	5.62
420.3	Endocarditis	280	359620	0.86 (0.84, 0.88)	<0.001	6.80
428.4	Heart failure with preserved EF [Diastolic heart failure]	1740	358160	0.86 (0.85, 0.86)	<0.001	6.77
428.1	Congestive heart failure (CHF) NOS	3040	356860	0.86 (0.85, 0.86)	<0.001	6.87
426.91	Cardiac pacemaker in situ	2120	357780	0.85 (0.85, 0.86)	<0.001	7.60
428.3	Heart failure with reduced EF [Systolic or combined heart failure]	2880	357020	0.85 (0.84, 0.86)	<0.001	8.08
440.2	Atherosclerosis of the extremities	640	359260	0.84 (0.83, 0.86)	<0.001	8.30
428	Congestive heart failure; nonhypertensive	6700	353200	0.84 (0.84, 0.85)	<0.001	6.00
426.92	Cardiac defibrillator in situ	1220	358680	0.84 (0.83, 0.85)	<0.001	14.68
394.7	Disease of tricuspid valve	440	359460	0.84 (0.82, 0.85)	<0.001	5.01
426.9	Cardiac pacemaker/device in situ	3440	356460	0.84 (0.83, 0.85)	<0.001	8.23
411.8	Other chronic ischemic heart disease, unspecified	480	359420	0.84 (0.82, 0.85)	<0.001	6.32
427.21	Atrial fibrillation	8340	351560	0.84 (0.83, 0.84)	<0.001	8.46
427.2	Atrial fibrillation and flutter	9260	350640	0.83 (0.83, 0.84)	<0.001	8.16
395.6	Heart valve replaced	1100	358800	0.83 (0.82, 0.84)	<0.001	6.18
425.8	Other cardiomyopathy	200	359700	0.83 (0.81, 0.85)	<0.001	4.12
394.2	Mitral valve disease	700	359200	0.83 (0.81, 0.84)	<0.001	5.40
395.2	Nonrheumatic aortic valve disorders	3360	356540	0.82 (0.81, 0.82)	<0.001	4.46
443.9	Peripheral vascular disease, unspecified	2520	357380	0.82 (0.81, 0.83)	<0.001	4.75
427.8	Sinoatrial node dysfunction (Bradycardia)	1340	358560	0.82 (0.81, 0.83)	<0.001	4.04
394	Rheumatic disease of the heart valves	1080	358820	0.81 (0.80, 0.83)	<0.001	5.46
395.3	Nonrheumatic tricuspid valve disorders	140	359760	0.81 (0.79, 0.84)	<0.001	3.65
427.22	Atrial flutter	1680	358220	0.81 (0.80, 0.82)	<0.001	7.95
433.1	Occlusion and stenosis of precerebral arteries	1520	358380	0.81 (0.80, 0.82)	<0.001	3.84
426.32	Left bundle branch block	1180	358720	0.80 (0.79, 0.81)	<0.001	3.93
427.12	Paroxysmal ventricular tachycardia	1520	358380	0.79 (0.78, 0.80)	<0.001	5.51
426.3	Bundle branch block	2080	357820	0.78 (0.78, 0.79)	<0.001	3.21
427.42	Cardiac arrest	440	359460	0.78 (0.76, 0.80)	<0.001	4.02
442.1	Aortic aneurysm	1900	358000	0.78 (0.77, 0.79)	<0.001	3.29
440	Atherosclerosis	2120	357780	0.78 (0.77, 0.79)	<0.001	3.96
427.4	Cardiac arrest and ventricular fibrillation	500	359400	0.78 (0.76, 0.79)	<0.001	3.05
395	Heart valve disorders	6520	353380	0.78 (0.77, 0.78)	<0.001	3.55
426.2	Atrioventricular [AV] block	1240	358660	0.77 (0.75, 0.78)	<0.001	6.37
428.2	Heart failure NOS	400	359500	0.77 (0.75, 0.79)	<0.001	3.11
433.11	Occlusion of cerebral arteries, with cerebral infarction	160	359740	0.76 (0.73, 0.80)	<0.001	2.91
411.4	Coronary atherosclerosis	9760	350140	0.76 (0.76, 0.77)	<0.001	2.66
426	Cardiac conduction disorders	7380	352520	0.76 (0.76, 0.77)	<0.001	4.15
425	Cardiomyopathy	3440	356460	0.76 (0.75, 0.77)	<0.001	5.67
440.1	Atherosclerosis of renal artery	240	359660	0.76 (0.73, 0.78)	<0.001	2.94

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Table 9 continued from previous page

Phecode	Description	N_{pos}	N_{neg}	AUC	P-Value	AUPRC Lift
411	Ischemic Heart Disease	12520	347380	0.76 (0.75, 0.76)	<0.001	2.63
411.1	Unstable angina (intermediate coronary syndrome)	540	359360	0.75 (0.73, 0.77)	<0.001	2.60
425.1	Primary/intrinsic cardiomyopathies	3120	356780	0.75 (0.74, 0.76)	<0.001	5.40
411.2	Myocardial infarction	4060	355840	0.75 (0.74, 0.75)	<0.001	2.73
457.2	Encounter for long-term (current) use of antiplatelets/antithrombotics	760	359140	0.75 (0.73, 0.76)	<0.001	2.99
443	Peripheral vascular disease	3860	356040	0.75 (0.74, 0.75)	<0.001	3.13
442.3	Aneurysm of artery of lower extremity	100	359800	0.75 (0.69, 0.80)	<0.001	68.74
433.21	Cerebral artery occlusion, with cerebral infarction	4480	355420	0.75 (0.74, 0.75)	<0.001	2.63
433.2	Occlusion of cerebral arteries	4500	355400	0.75 (0.74, 0.75)	<0.001	2.68
442.11	Abdominal aortic aneurysm	600	359300	0.74 (0.72, 0.76)	<0.001	2.93
457	Encounter for long-term (current) use of anticoagulants, antithrombotics, aspirin	900	359000	0.74 (0.73, 0.75)	<0.001	2.31
433	Cerebrovascular disease	10000	349900	0.74 (0.73, 0.74)	<0.001	2.52
433.3	Cerebral ischemia	3940	355960	0.74 (0.73, 0.74)	<0.001	2.82
411.9	Other acute and subacute forms of ischemic heart disease	440	359460	0.74 (0.72, 0.76)	<0.001	4.63
401	Hypertension	46820	313080	0.74 (0.73, 0.74)	<0.001	1.98
440.9	Atherosclerosis of aorta	1120	358780	0.73 (0.72, 0.75)	<0.001	2.20
429.2	Abnormal function study of cardiovascular system	340	359560	0.73 (0.71, 0.76)	<0.001	2.85
426.23	Second degree AV block	260	359640	0.73 (0.70, 0.77)	<0.001	3.05
401.1	Essential hypertension	46240	313660	0.73 (0.73, 0.74)	<0.001	1.97
433.31	Transient cerebral ischemia	3700	356200	0.73 (0.73, 0.74)	<0.001	2.46
447.1	Stricture of artery	100	359800	0.73 (0.68, 0.78)	<0.001	3.27
401.3	Other hypertensive complications	1120	358780	0.73 (0.72, 0.74)	<0.001	2.10
442	Other aneurysm	3040	356860	0.73 (0.72, 0.74)	<0.001	2.52
416	Cardiomegaly	1200	358700	0.73 (0.72, 0.74)	<0.001	2.20
426.31	Right bundle branch block	820	359080	0.73 (0.71, 0.74)	<0.001	2.36
401.2	Hypertensive heart and/or renal disease	880	359020	0.73 (0.71, 0.74)	<0.001	2.38
425.12	Other hypertrophic cardiomyopathy	500	359400	0.73 (0.70, 0.75)	<0.001	5.84
414	Other forms of chronic heart disease	840	359060	0.72 (0.71, 0.74)	<0.001	3.14
415.2	Chronic pulmonary heart disease	2240	357660	0.72 (0.71, 0.73)	<0.001	2.52
459.9	Circulatory disease NEC	3320	356580	0.72 (0.71, 0.73)	<0.001	2.63
411.3	Angina pectoris	1560	358340	0.72 (0.71, 0.73)	<0.001	1.99
395.1	Nonrheumatic mitral valve disorders	3200	356700	0.72 (0.71, 0.72)	<0.001	2.11
425.2	Secondary/extrinsic cardiomyopathies	180	359720	0.71 (0.67, 0.74)	<0.001	2.46
447	Other disorders of arteries and arterioles	1880	358020	0.71 (0.70, 0.72)	<0.001	2.21
458.9	Hypotension NOS	1800	358100	0.70 (0.68, 0.71)	<0.001	2.23
Chapter IX: Respiratory						
505	Other pulmonary inflammation or edema	520	359380	0.80 (0.79, 0.82)	<0.001	4.77
510.2	Lung transplant	880	359020	0.80 (0.79, 0.81)	<0.001	3.42
496	Chronic airway obstruction	4520	355380	0.78 (0.77, 0.78)	<0.001	3.03
510	Other diseases of lung	1420	358480	0.78 (0.77, 0.79)	<0.001	2.82
501	Pneumonitis due to inhalation of food or vomitus	1040	358860	0.78 (0.76, 0.79)	<0.001	5.76
504.1	Idiopathic fibrosing alveolitis	220	359680	0.77 (0.74, 0.79)	<0.001	2.73
503	Pulmonary congestion and hypostasis	100	359800	0.76 (0.72, 0.80)	<0.001	2.91
496.1	Emphysema	580	359320	0.76 (0.75, 0.78)	<0.001	3.23
504	Other alveolar and parietoalveolar pneumonopathy	1040	358860	0.76 (0.75, 0.77)	<0.001	2.41
509.1	Respiratory failure	3100	356800	0.75 (0.75, 0.76)	<0.001	3.18
496.2	Chronic bronchitis	1180	358720	0.75 (0.74, 0.76)	<0.001	2.63
496.3	Bronchiectasis	960	358940	0.74 (0.73, 0.75)	<0.001	2.27
496.21	Obstructive chronic bronchitis	840	359060	0.74 (0.72, 0.75)	<0.001	2.34
502	Postinflammatory pulmonary fibrosis	400	359500	0.73 (0.71, 0.75)	<0.001	2.14
480.5	Bronchopneumonia and lung abscess	160	359740	0.73 (0.70, 0.76)	<0.001	2.98
480.1	Bacterial pneumonia	900	359000	0.72 (0.70, 0.74)	<0.001	2.37
480	Pneumonia	7000	352900	0.71 (0.71, 0.72)	<0.001	2.25
509	Respiratory failure, insufficiency, arrest	5940	353960	0.71 (0.71, 0.72)	<0.001	2.25
507	Pleurisy; pleural effusion	2480	357420	0.71 (0.70, 0.72)	<0.001	2.48
500.1	Extrinsic allergic alveolitis	160	359740	0.70 (0.67, 0.73)	<0.001	1.70
480.11	Pneumococcal pneumonia	220	359680	0.70 (0.67, 0.73)	<0.001	2.20
509.2	Respiratory insufficiency	440	359460	0.70 (0.68, 0.72)	<0.001	2.32

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Phcode	Description	N_{pos}	N_{neg}	AUC	P-Value	AUPRC Lift
Chapter X: Digestive						
560.3	Peritoneal or intestinal adhesions	100	359800	0.79 (0.76, 0.82)	<0.001	2.77
556	Ulceration of the lower GI tract	180	359720	0.75 (0.72, 0.78)	<0.001	5.58
556.1	Ulceration of intestine	160	359740	0.74 (0.71, 0.77)	<0.001	3.09
571.81	Portal hypertension	960	358940	0.71 (0.70, 0.73)	<0.001	2.44
565.1	Anal and rectal polyp	120	359780	0.71 (0.67, 0.76)	<0.001	2.76
562.1	Diverticulosis	3440	356460	0.71 (0.70, 0.71)	<0.001	1.97
573.4	Acute and subacute necrosis of liver	400	359500	0.70 (0.67, 0.73)	<0.001	3.16
Chapter XI: Genitourinary						
618.5	Prolapse of vaginal vault after hysterectomy	100	359800	0.86 (0.84, 0.88)	<0.001	4.14
585.32	End stage renal disease	2660	357240	0.78 (0.78, 0.79)	<0.001	4.44
585.2	Renal failure NOS	360	359540	0.78 (0.76, 0.81)	<0.001	5.42
627.1	Postmenopausal bleeding	760	359140	0.78 (0.77, 0.79)	<0.001	2.96
585.3	Chronic renal failure [CKD]	6840	353060	0.75 (0.74, 0.75)	<0.001	2.87
618.1	Prolapse of vaginal walls	1620	358280	0.75 (0.74, 0.76)	<0.001	2.37
627.3	Postmenopausal atrophic vaginitis	1260	358640	0.74 (0.73, 0.75)	<0.001	2.60
600	Hyperplasia of prostate	5960	353940	0.74 (0.74, 0.75)	<0.001	2.65
585.31	Renal dialysis	1720	358180	0.74 (0.73, 0.75)	<0.001	3.15
588	Disorders resulting from impaired renal function	1180	358720	0.74 (0.73, 0.75)	<0.001	3.65
580.2	Nephrotic syndrome without mention of glomerulonephritis	540	359360	0.74 (0.72, 0.76)	<0.001	5.07
588.2	Secondary hyperparathyroidism (of renal origin)	1120	358780	0.74 (0.73, 0.75)	<0.001	3.54
585.34	Chronic Kidney Disease, Stage IV	1500	358400	0.74 (0.72, 0.75)	<0.001	2.90
585	Renal failure	14680	345220	0.73 (0.73, 0.74)	<0.001	2.53
618	Genital prolapse	2220	357680	0.73 (0.72, 0.74)	<0.001	2.50
627	Menopausal and postmenopausal disorders	3880	356020	0.72 (0.71, 0.73)	<0.001	2.27
585.1	Acute renal failure	8840	351060	0.71 (0.71, 0.72)	<0.001	2.35
599.6	Oliguria and anuria	140	359760	0.71 (0.68, 0.74)	<0.001	1.69
585.4	Chronic kidney disease, Stage I or II	1100	358800	0.71 (0.70, 0.72)	<0.001	2.80
611.1	Abnormal mammogram	1600	358300	0.71 (0.69, 0.72)	<0.001	2.21
627.2	Symptomatic menopause	1520	358380	0.71 (0.69, 0.72)	<0.001	2.29
587	Kidney replaced by transplant	2360	357540	0.71 (0.70, 0.72)	<0.001	2.70
796	Elevated prostate specific antigen [PSA]	1940	357960	0.70 (0.69, 0.72)	<0.001	2.90
599.4	Urinary incontinence	3860	356040	0.70 (0.70, 0.71)	<0.001	1.91
618.2	Uterine/Uterovaginal prolapse	880	359020	0.70 (0.69, 0.72)	<0.001	1.95
586.2	Cyst of kidney, acquired	1120	358780	0.70 (0.68, 0.71)	<0.001	2.31
Chapter XII: Pregnancy complications						
653	Problems associated with amniotic cavity and membranes	1180	358720	0.74 (0.72, 0.75)	<0.001	2.20
635.2	Antepartum hemorrhage, abruptio placentae, and placenta previa	2360	357540	0.72 (0.71, 0.73)	<0.001	2.12
647	Infectious and parasitic complications affecting pregnancy	600	359300	0.72 (0.70, 0.74)	<0.001	3.01
635	Hemorrhage during pregnancy; childbirth and postpartum	2600	357300	0.71 (0.70, 0.72)	<0.001	2.09
643	Excessive vomiting in pregnancy	2060	357840	0.71 (0.70, 0.72)	<0.001	1.91
636.3	Hemorrhage in early pregnancy	4320	355580	0.71 (0.70, 0.72)	<0.001	1.97
636	Early or threatened labor; hemorrhage in early pregnancy	4560	355340	0.71 (0.70, 0.71)	<0.001	1.86
635.3	Placenta previa and abruptio placenta	280	359620	0.70 (0.67, 0.73)	<0.001	2.45
Chapter XIII: Dermatologic						
701.1	Keratoderma, acquired	140	359760	0.78 (0.74, 0.82)	<0.001	4.64
702.4	Degenerative skin disorders	120	359780	0.78 (0.74, 0.81)	<0.001	5.26
707.2	Chronic ulcer of leg or foot	2060	357840	0.77 (0.76, 0.78)	<0.001	3.46
702.1	Actinic keratosis	1860	358040	0.75 (0.74, 0.76)	<0.001	2.94
707	Chronic ulcer of skin	3220	356680	0.73 (0.73, 0.74)	<0.001	2.85
938.2	Chronic dermatitis due to solar radiation	280	359620	0.73 (0.69, 0.76)	<0.001	3.94
702	Degenerative skin conditions and other dermatoses	2640	357260	0.72 (0.71, 0.73)	<0.001	2.39
709.3	Systemic sclerosis	260	359640	0.72 (0.69, 0.75)	<0.001	3.52
687.3	Changes in skin texture	120	359780	0.72 (0.68, 0.76)	<0.001	2.03
707.1	Decubitus ulcer	860	359040	0.72 (0.70, 0.73)	<0.001	2.61
709.6	Other specified diffuse diseases of connective tissue	200	359700	0.70 (0.66, 0.74)	<0.001	3.52

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Phcode	Description	N_{pos}	N_{neg}	AUC	P-Value	AUPRC Lift
Chapter XIV: Musculoskeletal						
717	Polymyalgia Rheumatica	640	359260	0.81 (0.79, 0.82)	<0.001	3.25
731.1	Osteitis deformans [Paget's disease of bone]	100	359800	0.79 (0.75, 0.82)	<0.001	3.31
743.11	Osteoporosis NOS	7700	352200	0.78 (0.78, 0.79)	<0.001	2.99
743.1	Osteoporosis	7700	352200	0.78 (0.77, 0.78)	<0.001	2.85
743	Osteoporosis, osteopenia and pathological fracture	9000	350900	0.76 (0.76, 0.77)	<0.001	2.81
740.2	Osteoarthritis, generalized	440	359460	0.76 (0.74, 0.77)	<0.001	2.15
737.1	Kyphosis (acquired)	260	359640	0.74 (0.71, 0.77)	<0.001	2.67
710.1	Osteomyelitis	2460	357440	0.73 (0.72, 0.74)	<0.001	3.25
740.9	Osteoarthritis NOS	4720	355180	0.73 (0.73, 0.74)	<0.001	2.27
740	Osteoarthritis	14640	345260	0.73 (0.73, 0.73)	<0.001	2.14
710.11	Acute osteomyelitis	480	359420	0.73 (0.71, 0.75)	<0.001	4.66
720.1	Spinal stenosis of lumbar region	3960	355940	0.73 (0.72, 0.74)	<0.001	2.28
710	Osteomyelitis, periostitis, and other infections involving bone	2460	357440	0.73 (0.72, 0.74)	<0.001	3.32
710.19	Unspecified osteomyelitis	1980	357920	0.73 (0.72, 0.74)	<0.001	2.61
740.11	Osteoarthritis, localized, primary	10360	349540	0.73 (0.72, 0.73)	<0.001	2.09
740.1	Osteoarthritis; localized	11100	348800	0.72 (0.72, 0.73)	<0.001	2.06
727.5	Rupture of synovium	560	359340	0.72 (0.70, 0.73)	<0.001	2.06
735.21	Hammer toe (acquired)	300	359600	0.71 (0.69, 0.74)	<0.001	5.38
720	Spinal stenosis	6100	353800	0.71 (0.71, 0.72)	<0.001	2.20
739	Contracture of joint	440	359460	0.71 (0.68, 0.73)	<0.001	2.97
731	Osteitis deformans and osteopathies associated with other disorders classified elsewhere	180	359720	0.70 (0.67, 0.73)	<0.001	2.44
735.2	Acquired toe deformities	620	359280	0.70 (0.68, 0.72)	<0.001	2.10
721.2	Spondylosis with myelopathy	540	359360	0.70 (0.68, 0.71)	<0.001	1.79
Chapter XVI: Symptoms						
797.1	Cardiogenic shock	260	359640	0.89 (0.88, 0.90)	<0.001	14.89
791	Gangrene	300	359600	0.83 (0.81, 0.85)	<0.001	14.41
797	Shock	580	359320	0.79 (0.77, 0.81)	<0.001	6.51
795.8	Abnormal tumor markers	120	359780	0.75 (0.71, 0.79)	<0.001	7.55
772.3	Muscle weakness	2240	357660	0.70 (0.69, 0.71)	<0.001	2.37
Chapter XVII: Injuries & poisonings						
994.21	Septic shock	480	359420	0.87 (0.85, 0.88)	<0.001	10.73
874	Complication of amputation stump	100	359800	0.82 (0.78, 0.86)	<0.001	24.32
961	Poisoning by other anti-infectives	120	359780	0.81 (0.77, 0.84)	<0.001	4.31
854	Complications of cardiac/vascular device, implant, and graft	1220	358680	0.76 (0.75, 0.78)	<0.001	4.53
800.1	Fracture of neck of femur	1580	358320	0.75 (0.74, 0.76)	<0.001	2.51
972	Poisoning by agents primarily affecting the cardiovascular system	100	359800	0.72 (0.67, 0.77)	<0.001	2.49
994.2	Sepsis	5660	354240	0.72 (0.71, 0.72)	<0.001	2.89
994	Sepsis and SIRS	6020	353880	0.71 (0.71, 0.72)	<0.001	2.64
851	Complications of transplants and reattached limbs	2000	357900	0.71 (0.70, 0.72)	<0.001	2.30
816	Cerebral laceration and contusion	120	359780	0.70 (0.65, 0.76)	<0.001	2.49
858	Complication of internal orthopedic device	1560	358340	0.70 (0.69, 0.71)	<0.001	2.00

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